

GIAO ^{13}C NMR Calculation with Sorted Training Sets Improves Accuracy and Reliability for Structural Assignment

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Cite This: *J. Org. Chem.* 2020, 85, 11350–11358

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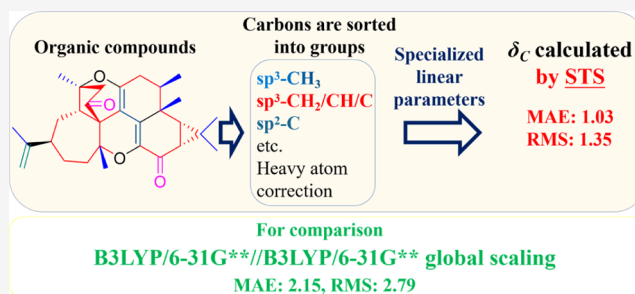


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ABSTRACT: GIAO ^{13}C NMR calculation plays important roles in structural assignment for small organic molecules. Especially, higher accuracy and confidence are required for the structural assignment of complex organic molecules. In this GIAO ^{13}C NMR calculation strategy, carbons were sorted according to their type of hybridization, radii of solvation cavity, or solvent interactions. The calculated shielding tensors of carbons in each sorted training set were linear-regressed with experimental data separately, and the obtained linear parameters were used to convert calculated shielding tensors into calculated chemical shifts. This approach shows significantly improved accuracy, especially for sp^2 carbons, compared to conventional GIAO ^{13}C NMR calculation protocols. A statistic-based probability algorithm was proposed to quantify the reliability of structural assignment. With empirical linear parameters for three commonly used NMR solvents as well as an easy-to-use script and a spreadsheet, this ^{13}C NMR calculation protocol provides a useful tool for structural validation or assignment of ambiguous organic structures.



INTRODUCTION

NMR calculation has become a powerful tool to settle many tough tasks of structural assignment of natural or synthetic organic molecules, which are difficult to be solved by conventional structural elucidation methods (1D, 2D NMR, MS, ECD etc.).^{1–7} Especially, gauge invariant atomic orbital (GIAO) NMR calculation is the most prevalent method applied to constitute a molecular structure, assign relative configurations, validate structural assignments, and revise the structures wrongly assigned.^{1–7} In previous studies, many efforts were spent on achieving accurate theoretic ^{13}C NMR calculation.^{5,6} For instance, the level of theory and basis sets for ^{13}C NMR calculation,^{8–13} as well as the relationship between calculated shielding tensors and experimental data, have been extensively investigated.^{14,15} For the data processing of calculated shielding tensors, methods including MSTD,¹⁶ scaling correction,⁶ empirical scaling from data sets,⁶ and correction for certain chemical groups^{4,17} were proposed to obtain more accurately calculated chemical shifts. Moreover, probability or data quality measuring methods including CP3,¹⁸ DP4,³ DP4+,¹⁹ KDE,²⁰ DiCE,²¹ and DU8+⁴ were introduced to evaluate the validation of calculated data of assigned structures, which were widely used in structural assignment with tremendous success.²²

Compared to ^1H NMR calculation, ^{13}C NMR calculation is more frequently used because of higher accuracy of calculation and less sensitivity to the interfering factors in NMR measuring (instruments, solvents, temperature, sample concentration, etc.).⁶

The calculated shielding tensors and chemical shifts generally have a linear relationship, which can be described with a linear equation $\delta = \mathbf{aI} + \mathbf{b}$ (δ is chemical shifts, and I is calculated shielding tensors).⁶ Based on this important property, empirical scaling methods, obtaining the values of parameters $\mathbf{a}$ (slope) and $\mathbf{b}$ (intercept) from a set of training molecules, are capable of cancelling many systemic errors with low computational cost, which are superior to using a few molecules as ref 6. For instance, Tantillo et al. composed the CHESHIRE repository covering linear parameters for many commonly used NMR solvents, geometry optimization, and GIAO NMR calculation methods, which provided considerable convenience for many NMR calculation-involved research studies (<http://cheshirenmr.info>). Moreover, Reddy and Kutateladeze reported the global linear parameters regressed with experimental data measured in CDCl_3 and shielding tensors calculated on $\omega\text{B97X-D}/6\text{-}31\text{G}^*/\text{B3LYP}/6\text{-}31\text{G}^*$ (gas phase) level of theory, showing good R^2 value (0.999).⁴ These methods have satisfactory accuracy for most sp^3 carbons and several type of sp^2 carbons.

Received: June 19, 2020

Published: August 3, 2020



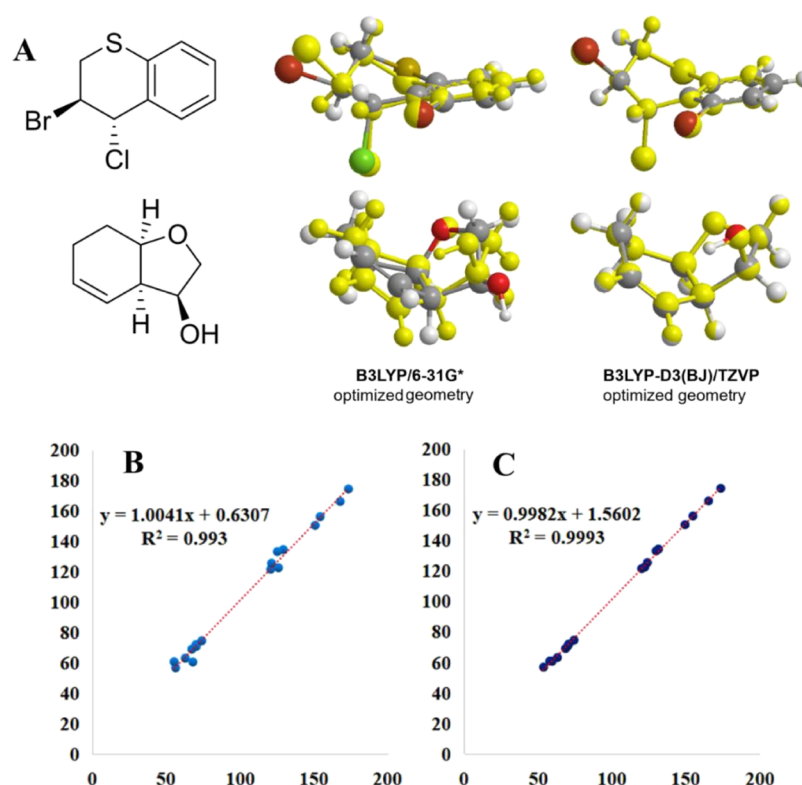


Figure 1. Two artificially constructed structures for optimization method evaluation. (A) Structures optimized on B3LYP/6-31G* and B3LYP-D3(BJ)/TZVP level of theory (gas phase) with ω B97X-D/def2-TZVPD optimized geometries as references (the yellow colored frames), (B) Shielding tensors of these two structures calculated on ω B97X-D/6-31G*//B3LYP/6-31G* level (x axis) regressed with the ones calculated on ω B97X-D/6-31G*// ω B97X-D/def2-TZVPD level (y axis), C. The shielding tensors of these two structures calculated on ω B97X-D/6-31G*//B3LYP-D3(BJ)/TZVP level (x axis) regressed with the ones calculated on ω B97X-D/6-31G*// ω B97X-D/def2-TZVPD level (y axis).

Accuracy is the key issue for ^{13}C NMR calculation, representing the resolving power of methods, no matter which probability evaluation methods are utilized. Because errors inevitably exist in every calculation, some incorrect structures may fortuitously fit the experimental data better than the correct one, which possesses the risk of misleading conclusions. The sources of error in theoretic calculation have been extensively investigated in previous research studies, such as electron correlation,²³ influence of solvent molecules,^{24,25} conformational flexibility,^{24,25} rovibration effects,²⁶ heavy-atom light-atom (HALA) effect,²⁷ etc. For heavy atom (Cl, S, Br, I, etc.) bearing carbons, several empirical correction methods were proposed to cancel the errors introduced by the HALA effect.^{4,7,17,28} However, some types of carbons including alkyne, carboxyl, and terminal double bond ($\text{sp}^2\text{-CH}_2$) show relatively higher errors with global linear parameters. For this issue, Xin et al. proposed an empirical correction of -4.0 ppm for carboxyl groups,¹⁷ and Andrews and Spivey used fragment referencing to cancel systematic errors for certain chemical groups.²⁸

To achieve higher accuracy, herein, we developed a new GIAO ^{13}C NMR calculation protocol to overcome the shortages of current methods. In this protocol, carbons are sorted by their solvation radii, hybridization types, and substitution groups, and the linear parameters for each group are trained individually. This so-called sorted training set strategy (STS) cancels the errors caused by the discrepancies between types of carbons.

On the other hand, in NMR calculation, some carbons with high deviations caused discrepancy in the tail region of

error distribution, compared to the assumed Gaussian distribution.²⁰ Correction of carbons with high deviation will enhance the confidence for many statistic models. Moreover, because the number and type of carbons vary from case to case and different types of carbon have different systematic errors, it is an advantage that the STS strategy has error distribution data for specific types of carbons and can provide normalized and unified statistic evaluation for diverse compounds. Herein, we have also proposed a statistic model for probability calculation ($P_{\text{mean}}/P_{\text{rel}}$) compatible with the performance of the STS strategy, on the basis of the sDP4+ algorithm.¹⁹

RESULTS AND DISCUSSION

For main group elements, it is widely accepted that the commonly used DFT functionals (e.g., B3LYP) combined with popular double zeta basis sets (e.g., 6-31G*) are sufficient for geometry optimization for NMR calculation.⁶ We artificially constructed a test group of small organic molecules, which covers bulky, aromatic, highly conjugated, and heavy atom-substituted structures (Figure S2). The shielding tensors of those structures calculated from the geometries optimized on B3LYP/6-31G* level of theory have a splendid linear relationship with those from the geometries optimized on ω B97X-D/def2-TZVPD level (Figure S2). With scaling correction, not surprisingly, an entry-level of geometry optimization method like B3LYP/6-31G* can produce good results comparable to much more expensive methods in many cases. For the structures having nonbonding interactions, however, this level may not yield the correct optimal

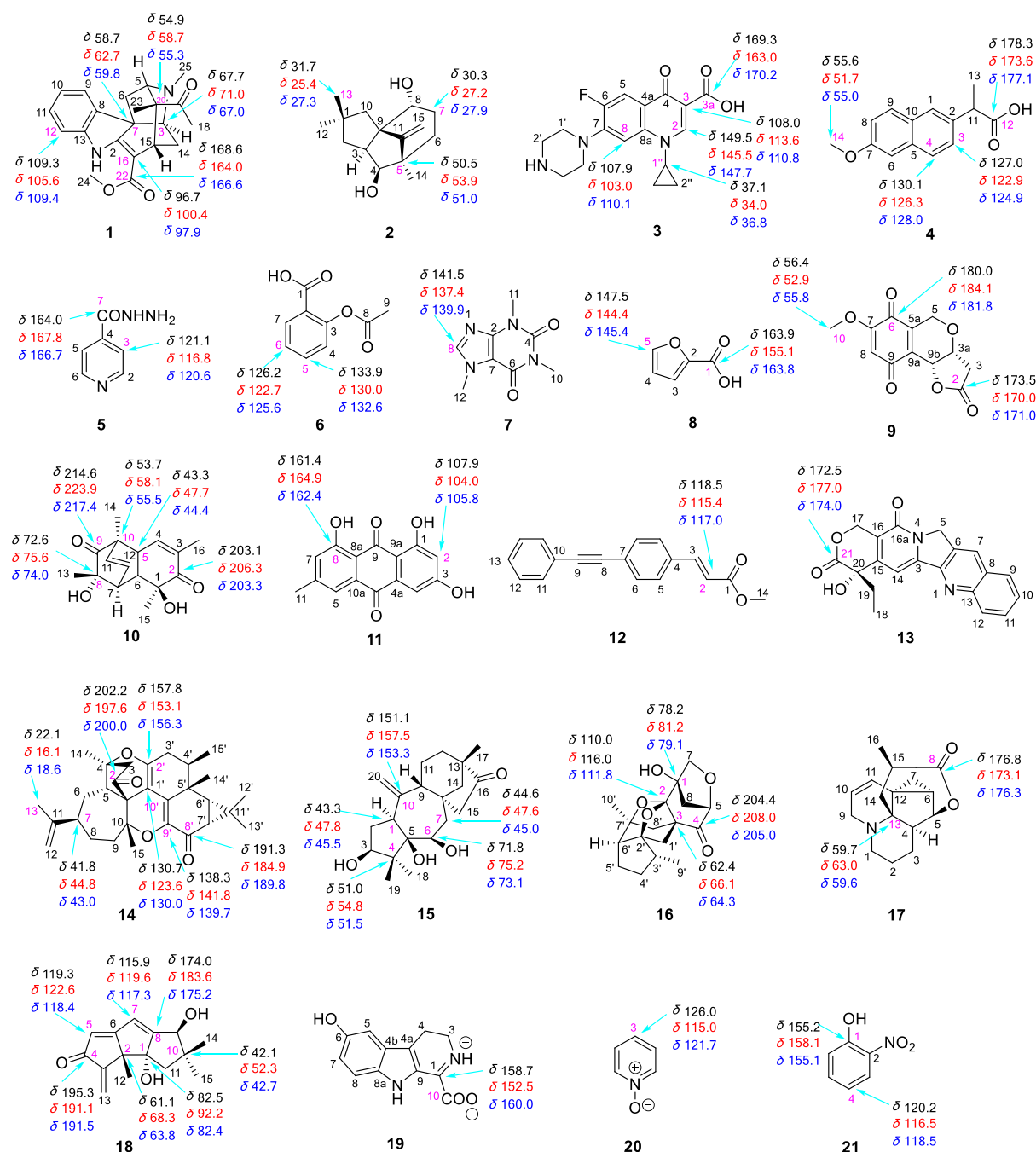


Figure 2. Test structures for benchmark and “difficult carbons” marked with experimental (black δ) and calculated ^{13}C NMR chemical shifts (red δ : by GSC, B3LYP/6-31G**//B3LYP/6-31G** level, gas phase; blue δ : by STS, $\omega\text{B97x-D}/6\text{-31G}^*/\text{B3LYP-D3(BJ)}/\text{TZVP}$ level, IEFPCM).

geometries in some cases (Figure 1), leading to large deviation of calculated shielding tensors compared to the reference method ($\omega\text{B97X-D}/\text{def2-TZVPD}$) (Figure 1). Therefore, we adopt B3LYP-D3(BJ)/TZVP (IEFPCM) as the geometry optimization level combined with GIAO NMR calculation on $\omega\text{B97X-D}/6\text{-31G}^*$ level for our protocol, which is a compromise between computational cost and the applicability for as many organic structures as possible.

In this protocol, we divide carbons by types of hybridization, radii of solvation cavity, or solvent interactions, and obtain their manner of linearity between calculated shielding tensors and experimental data, namely the parameters a and b of linear equation $\delta = aI + b$ trained from a set of molecules

(Figures 2 and S1). They were finally sorted into ten groups, namely $\text{sp}^3\text{-CH}_3$, $\text{sp}^3\text{-CH}_2/\text{CH}/\text{C}$, $\text{sp}^2\text{-CH}_2$, $\text{sp}^2\text{-CH}$ (except carbonyl group), $\text{sp}^2\text{-C}$ (except carbonyl group), X-C=O (carbonyl groups except carboxyl group, ketone, and aldehyde), COOH (carboxyl group), C=O (ketone and aldehyde), sp-CH , and sp-C , according to the similarity of linear parameters. The calculated chemical shifts of certain types of carbons were converted by corresponding linear parameters from calculated shielding tensors.

With mainstream GIAO ^{13}C NMR methods, some so-called “difficult carbons” have considerably larger errors than other carbons (e.g., carboxyl group). Therefore, we made a test set including representative classical drugs, drug

candidates, and natural products with “difficult carbons” to benchmark the performance of the newly proposed STS method (Figure 2). The experimental ^{13}C NMR data of the compounds in the benchmark set were recorded in CDCl_3 , CD_3OD , or $\text{DMSO}-d_6$.

For comparison, we performed GIAO NMR calculation on B3LYP/6-31G**//B3LYP/6-31G** (gas phase) level and RI-MP2/def2-SVP//B3LYP-D3(BJ)/TZVP (gas phase) level with global scaling correction (GSC), with linear parameters exactly trained by this test set. Because the performance of B3LYP/6-31G**//B3LYP/6-31G** level is significantly better than RI-MP2/def2-SVP//B3LYP-D3(BJ)/TZVP (gas phase) level, we adopt the former theoretic level with GSC as a comparison method herein. With scaling correction, notably, a common DFT functional like B3LYP showed better results than MP2 (using the calculated and experimental shielding tensor data reported by Frisch's group,²⁹ Table S1), which is consistent with the findings in our research. A benchmark showed that MP2 outperformed DFT methods,⁸ but it used CCSD(T) as references.

To eliminate the suspicion of deliberately manufacturing better results with limited number of samples, we combined another group of compounds (Figure S1) with the test set (Figure 2) to generate sorted training sets. The additional group of compounds have experimental ^{13}C NMR data reported from the literature,^{30,31} SDBS (<https://sdb.sdb.aist.go.jp>), or measured by ourselves in CDCl_3 , CD_3OD , or $\text{DMSO}-d_6$.

As a result, different groups of carbons showed more-or-less different linear parameters in the three solvents (Figure 3 and Table 1). In CDCl_3 , the linearity of each group is similar, except that the shielding tensors of COOH are significantly overestimated. In CD_3OD and DMSO , the shielding tensors of group $\text{sp}^2\text{-CH}_2$ are underestimated. Besides, in DMSO , the difference between the tendencies of sp^2 carbons and sp^3 carbons is higher than the other two solvents. These phenomena probably resulted from their different interaction activity with solvent molecules, which can hardly be accurately described by common implicit solvent models. This difference between carbons is a source of error of global scaling correction. By contrast, the STS strategy can partially cancel this kind of error, allowing more accurate calculation.

For the compounds in the test set (Figure 2), we first used the STS linear parameters to convert calculated shielding tensors into calculated chemical shifts, and then, they were globally scaling corrected to generate the final calculated chemical shifts. The overall mean absolute error (MAE) and root mean square (rms) values of the STS strategy are as low as 1.03 and 1.35 ppm, respectively, which are much better than those of GSC (B3LYP/6-31G**//B3LYP/6-31G** (gas phase) level) (Figure 4, Table 2), implying that the STS strategy is an excellent ^{13}C NMR calculation method.

Moreover, the STS strategy has extensibility to adapt to unusual situations. Researchers can readily use new training sets to add/adjust corrections for some unique carbons, if the experimental data of known similar structures are available.

First, the experimental chemical shifts of the carbons for further correction in a set of known structures need to be retroconverted into pseudo experimental shielding tensors (I_{retro}) by the corresponding linear parameters **a** and **b** in Table 1 (eq 1) and then regressed with calculated shielding tensors (I_{calcd}) to obtain the correction linear equation (eq 2).

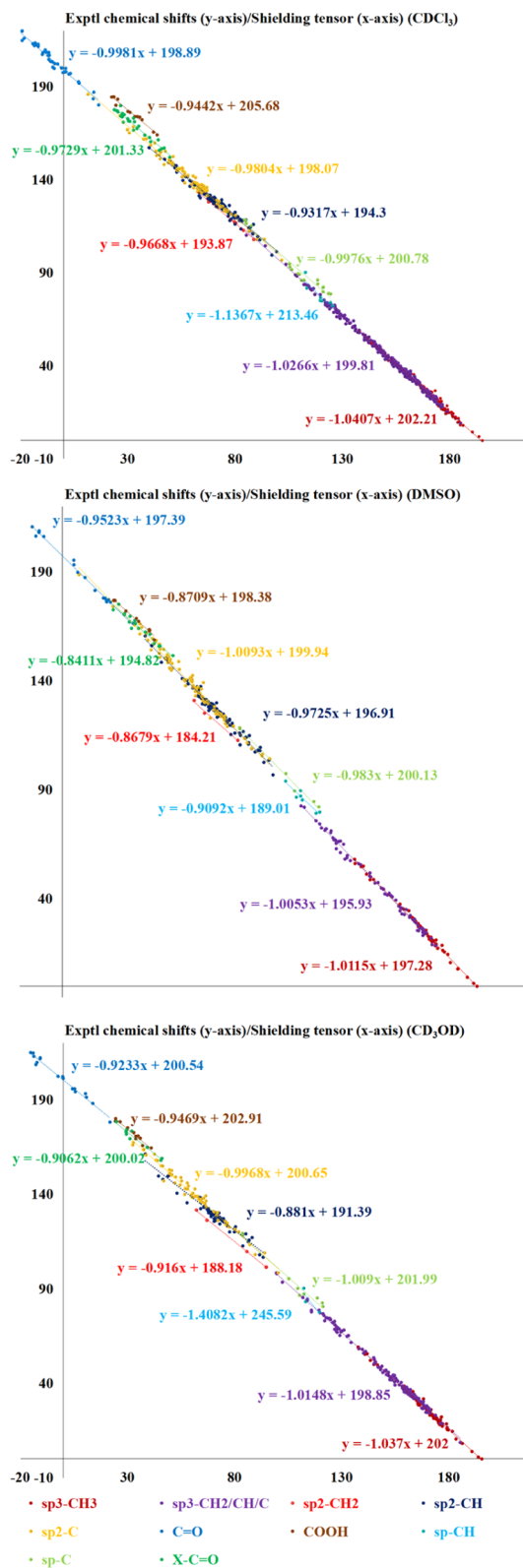


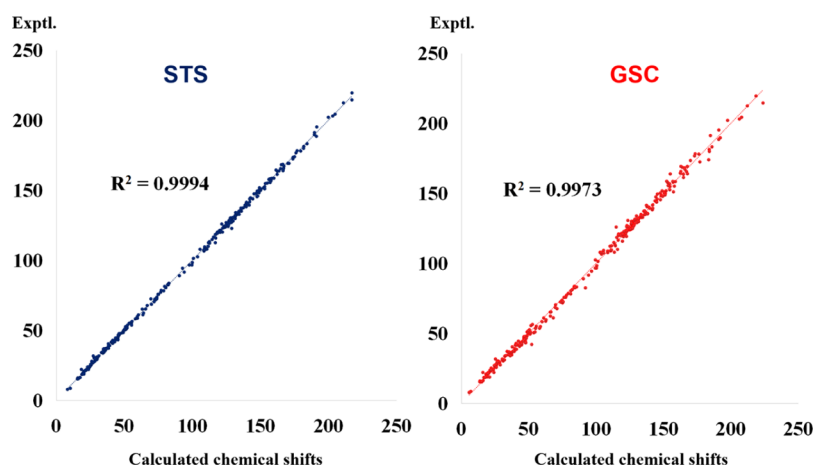
Figure 3. Linear regression of each carbon group in the training sets between experimental chemical shifts (y axis) and calculated shielding tensors (x axis) in three solvents, respectively (CDCl_3 , $\text{DMSO}-d_6$, and CD_3OD).

$$I_{\text{retro}} = (\delta_{\text{exptl}} - \mathbf{b})/\mathbf{a} \quad (1)$$

$$I_{\text{retro}} = \mathbf{a}I_{\text{calcd}} + \mathbf{b} \quad (2)$$

Table 1. Values of Linear Parameters *a* and *b* for the Linear Equation $\delta = aI + b$ Calculated from the Training Sets in Three Solvents with the STS Strategy

carbon groups	CDCl ₃		DMSO- <i>d</i> ₆		CD ₃ OD	
	<i>a</i>	<i>b</i>	<i>a</i>	<i>b</i>	<i>a</i>	<i>b</i>
sp ³ -CH ₃	−1.0407	202.21	−1.0115	197.28	−1.0370	202.00
sp ³ -CH ₂ /CH/C	−1.0266	199.81	−1.0053	195.93	−1.0148	198.85
sp ² -CH ₂	−0.9668	193.87	−0.8679	184.21	−0.9160	188.18
sp ² -CH	−0.9317	194.30	−0.9725	196.91	−0.8810	191.39
sp ² -C	−0.9804	198.07	−1.0093	199.94	−0.9968	200.65
C=O	−0.9981	198.89	−0.9523	197.39	−0.9233	200.54
X-C=O	−0.9729	201.33	−0.8411	194.82	−0.9062	200.02
COOH	−0.9442	205.68	−0.8709	198.38	−0.9469	202.91
sp-CH	−1.1367	213.46	−0.9092	189.01	−1.4082	245.59
sp-C	−0.9976	200.78	−0.9830	200.13	−1.0090	201.99

**Figure 4.** Linear regression of the experimental chemical shifts and calculated chemical shifts by STS (ω B97x-D/6-31G**//B3LYP-D3(BJ)/TZVP level, IEFPCM) and GSC (B3LYP/6-31G**//B3LYP/6-31G** level, gas phase).**Table 2.** Comparison of the MAE and rms of Chemical Shifts Calculated by STS (ω B97x-D/6-31G**//B3LYP-D3(BJ)/TZVP Level, IEFPCM) and GSC (B3LYP/6-31G**//B3LYP/6-31G** Level, Gas Phase)

groups	STS	GSC	STS	GSC
	MAE (in ppm)	MAE (in ppm)	rms (in ppm)	rms (in ppm)
all	1.03	2.15	1.35	2.79
“difficult carbons” ^a	1.43	4.64	1.74	5.01
sp ³ -CH ₃	0.73	1.99	1.12	2.45
sp ³ -CH ₂ /CH/C	0.85	1.96	1.07	2.58
sp ² -CH ₂	0.65	1.29	0.68	1.79
sp ² -CH	1.16	2.16	1.50	2.76
sp ² -C	1.23	1.92	1.58	2.61
C=O	1.57	3.46	1.84	4.03
X-C=O	0.91	2.90	1.14	3.08
COOH	1.00	5.66	1.12	6.00
sp-C	1.97	2.30	2.03	2.38

^aDifficult carbons are the carbons with absolute error values more than 3 ppm, calculated by GSC on B3LYP/6-31G**//B3LYP/6-31G** level of theory.

In data processing, the calculated shielding tensors of the carbons needing further correction shall be corrected by eq 2 and then converted into calculated chemical shifts as regular carbons.

With the experimental data in the training set, herein, we demonstrate how to add further corrections for chlorine-, bromine-, and sulfur-bearing carbons, as well as sulfonates (Figure 5).

There are many methods to evaluate how well the calculated ¹³C NMR chemical shifts fit with the experimental data. But most of them can only tell us which calculated data are “better” in a group of possible structures. The reason is that different compounds have different types of carbons, and using the overall standard as a threshold will be too strict/loose for these compounds with more/less “difficult carbons”. In a circumstance that the correct structure may not be included in the proposed possible structures, a unified criterion is important. The STS strategy not only provides specialized correction for different types of carbons but also yields specialized statistics for each of them. Therefore, we can normalize the errors of different types of carbons, allowing us to calculate the overall distribution of errors and evaluate the data of most compounds with uniformed standards. With the idea like the probability assay reported by Sarotti’s group,¹⁹ we proposed the formula to calculate the probability of correct structural designation by ¹³C NMR calculation (eq 3). Under the assumption that the putative structure is correct, $e_{s,k}$ is the absolute error of the STS calculation of each carbon after scaling correction, and T^ν is the cumulative t distribution function with ν degrees of freedom, and σ_{STS} is the standard deviation of a certain group of carbons (sp³-CH₂/CH/C, sp²-CH, etc.) calculated by the

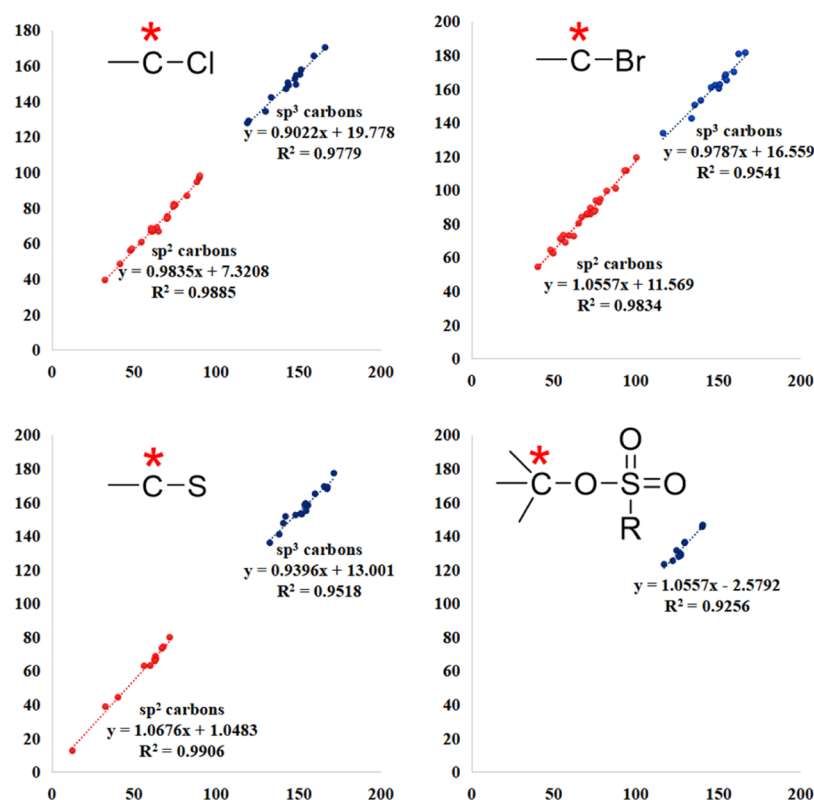


Figure 5. Linear relationship between I_{retro} (y axis) and I_{calcd} (x axis) of carbons needing further correction (red asterisk marked).

STS strategy. P_{mean} is the doubled value of the geometry average of the $1 - T^\nu(e_{s,k}/\sigma_{\text{STS}})$ values of all carbons in one structure. There is no widely accepted criterion that how large error (MAE or rms) denies a presumed structure. By contrast, P_{mean} provides a quantitative confidence to evaluate how possible a putative structure is wrongly proposed. Generally, if the P_{mean} value is more than 5%, with 95% confidence interval, the assumed structure can be accepted as a possible candidate, which is useful in checking whether the assigned structure is correct or verifying whether this computational strategy is suitable for the objective structures.

$$P_{\text{mean}} = 2 \times \sqrt[n]{\prod_{k=1}^n [1 - T^\nu(e_{s,k}/\sigma_{\text{STS}})]} \quad (3)$$

When there is more than one presumed structure with $P_{\text{mean}} > 5\%$, a relative probability evaluation like DP4³ or DP4+¹⁹ can be used to determine the correct structure from a group of candidates. On the basis of P_{mean} , thereby, we have proposed a relative probability (P_{rel}) formula as shown in eq 4, which is an analogue of sDP4+.¹⁹ Because of their similarity, the performance of P_{rel} should be close to sDP4+.

$$P_{\text{rel}} = \frac{\prod_{k=1}^n [1 - T^\nu(e_{s,k}^i/\sigma_{\text{STS}})]}{\sum_{j=1}^m \prod_{k=1}^n [1 - T^\nu(e_{s,k}^j/\sigma_{\text{STS}})]} \quad (4)$$

Herein, we use xylarichalasin A and 6-*epi*-xylarichalasin A³² (Figure 6) as a case study to show the performance of the STS strategy and probability calculation (Table 3). The MAE and rms values of the correct structure calculated by the STS strategy are much better than those calculated by the method utilized in the references (ω B97X-D/6-31G*/B3LYP/6-31G*, gas phase, and global scaling),^{4,32} which used the

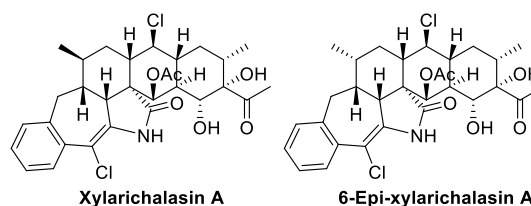


Figure 6. Structures of xylarichalasin A and 6-*epi*-xylarichalasin A.

same level for GIAO NMR calculation and introduced heavy atom correction as well. The improvement of accuracy is achieved by higher level of theory for optimization and specialized linear parameter from different types of carbons.

CONCLUSIONS

In summary, we have developed a GIAO ^{13}C NMR calculation protocol (STS) with significantly improved accuracy. Compared to the regularly used scaling correction method, the STS reduces over 50% value of overall MAE and rms and fits the experimental data much better for those “difficult carbons”. It is parameterized for three commonly used NMR solvents and is easy to be mended for unique cases. Besides, we have developed a statistic-based probability algorithm to estimate the probability of correct structural assignment, providing solid confidence to deduce unknown structures by GIAO ^{13}C NMR calculation. Moreover, we have integrated all the parameters and algorithms into one spreadsheet, which can be a handy tool for laboratory research routine. All the data of sorted training sets have been provided as Excel files in Supporting Information, and the users can readily modify and expand the data sets to adapt to their own research studies.

Table 3. GIAO ^{13}C NMR Calculations for Xylarichalasin A and 6-Epi-xylarichalasin A Using the STS Method (Chemical Shifts in ppm)

no.	exptl. ^a	calcd. xylarichalasin A	dev.	calcd. 6-epi-xylarichalasin A	dev.
1	176.2	176.0	0.2	175.9	0.3
3	137.9	138.6	0.7	139.1	1.2
4	45.0	45.1	0.1	44.1	0.9
5	50.1	50.2	0.1	47.9	2.2
6	30.7	30.4	0.3	28.8	1.9
7	33.6	33.5	0.1	32.8	0.8
8	37.2	37.6	0.4	38.7	1.5
9	54.3	55.6	1.3	55.9	1.6
10	103.2	104.3	1.1	104.3	1.1
11	37.4	37.1	0.3	35.5	1.9
12	20.1	19.8	0.3	20.7	0.6
13	67.4	68.6	1.2	68.6	1.2
14	42.0	40.7	1.3	41.1	0.9
15	32.8	32.8	0.0	33.6	0.8
16	35.9	37.0	1.1	37.5	1.6
17	83.5	84.6	1.1	84.8	1.3
18	212.8	212.1	0.7	211.8	1.0
19	72.7	71.6	1.1	71.9	0.8
20	41.7	42.9	1.2	43.2	1.5
21	70.5	69.4	1.1	69.7	0.8
22	25.7	23.5	2.2	24.1	1.6
23	14.7	15.2	0.5	15.9	1.2
24	136.7	136.2	0.5	136.6	0.1
25	135.4	138.1	2.7	138.1	2.7
26	129.1	128.2	0.9	128.1	1.1
27	127.5	127.0	0.6	126.8	0.7
28	127.0	125.9	1.1	125.9	1.1
29	127.7	127.9	0.2	127.8	0.1
30	171.2	170.6	0.6	170.7	0.5
31	20.5	20.1	0.4	20.6	0.1
		MAE	0.77	MAE	1.09
		rms	0.98	rms	1.24
		P_{mean}	49.9%	P_{mean}	33.9%
		P_{rel}	100.0%	P_{rel}	0.0%
		DP4 ^b	96.8%	DP4 ^b	3.2%
		MAE in ref 33	1.19	MAE in ref 33	1.44
		rms in ref 33	1.48	rms in ref 33	1.67
		sDP4+ in ref 33	99.8%	sDP4+ in ref 33	0.2%

^aData recorded in CDCl_3 (chemical shifts in ppm). ^bCalculated on B3LYP/6-31G**//MMFF level as described in the literature.³

EXPERIMENTAL METHODS

All the DFT calculations were performed by the Gaussian09 D01 software package or Gaussian16 software package with the *g09defaults* keyword.³³ For the flexible compounds, Confab was used to search the stable conformers with the torsion-driving approach.³⁴ The contribution of conformers was weighted by Boltzmann distribution based on Gibbs free energy calculated on B3LYP-D3(BJ)/TZVP (IEFPCM) level of theory. RI-MP2 NMR calculations were performed by the ORCA 4.1 software package.³⁵ The ^{13}C NMR of compounds 26, 27, 30, 57, 63, 66, 67, 68, and 69 in the additional training sets was measured on a Bruker AV-600 spectrometer (Bruker, Karlsruhe, Germany) in solvents CDCl_3 , CD_3OD , and $\text{DMSO}-d_6$ (150 MHz, 100 number of scans, pulse width value of 10.25, acquisition time value of 0.9044, and spectral width value of 36231.9). The chemical shifts of deuterated solvents were set as the reference (CDCl_3 77.06 ppm, CD_3OD 49.03 ppm, $\text{DMSO}-d_6$ 39.53 ppm).

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.joc.0c01451>.

Instructions for using the STS Excel file (*STS.xlsx*), python script (*extraction.py*), and computational data associated with this paper (*PDF*)

Experimental chemical shifts and shielding tensors calculated by the STS strategy in Excel files (*sorted_training_set-CDCl3.xlsx*, *sorted_training_set-CD3OD.xlsx*, and *sorted_training_set-DMSO.xlsx*); experimental chemical shifts and shielding tensors calculated on B3LYP/6-31G**//B3LYP/6-31G** (gas phase) and RI-MP2/def2-SVP//B3LYP-D3(BJ)/TZVP (gas phase) in the Excel file *B3LYP-RI-MP2_GSC.xlsx*; and experimental chemical shifts and shielding tensors of carbons with further correction (Cl-/Br-/S-bearing carbons and sulfonate) in the Excel file *heavy_atom_correction.xlsx* (*ZIP*)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This research was financially supported by the National Natural Science Foundation of China (31801789, 81903513, and 2191101099). We are grateful for the convenience for the NMR measurement provided by the Analytical & Measuring Center, School of Pharmaceutical Sciences, South-Central University for Nationalities. We are grateful for the assistance provided by Dr Kun Hu (Kunming Institute of Botany, Chinese Academy of Sciences).

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